

Question 1

```
MariaDB [(none)]> CREATE database EliteVideo;  
Query OK, 1 row affected (0.009 sec)
```

```
MariaDB [(none)]> use EliteVideo;  
Database changed  
MariaDB [EliteVideo]> create table PRICE (PRICE_CODE INT PRIMARY KEY,  
PRICE_DESCRIPTION VARCHAR(50) NOT NULL,  
PRICE_RENTFEE DECIMAL(5,2) NOT NULL,  
PRICE_DAILYLATEFEE DECIMAL(5,2) NOT NULL);  
Query OK, 0 rows affected (0.065 sec)
```

```
MariaDB [EliteVideo]> CREATE TABLE MOVIE (MOVIE_NUM INT PRIMARY KEY,  
MOVIE_TITLE VARCHAR(100) NOT NULL,  
MOVIE_YEAR INT NOT NULL,  
MOVIE_COST DECIMAL(6,2) NOT NULL,  
MOVIE_GENRE VARCHAR(50),  
PRICE_CODE INT NOT NULL,  
FOREIGN KEY (PRICE_CODE) REFERENCES PRICE(PRICE_CODE));  
Query OK, 0 rows affected (0.045 sec)
```

```
MariaDB [EliteVideo]> CREATE TABLE VIDEO (VID_NUM INT PRIMARY KEY,  
VID_INDATE DATE NOT NULL, MOVIE_NUM INT NOT NULL,  
FOREIGN KEY (MOVIE_NUM) REFERENCES MOVIE (MOVIE_NUM));  
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [EliteVideo]> CREATE TABLE MEMBERSHIP (MEM_NUM INT PRIMARY KEY,  
MEM_FNAME VARCHAR(50) NOT NULL,  
MEM_LNAME VARCHAR(50) NOT NULL,  
MEM_STREET VARCHAR(100) NOT NULL,  
MEM_CITY VARCHAR(50) NOT NULL,  
MEM_STATE VARCHAR(20) NOT NULL,  
MEM_ZIP VARCHAR(10) NOT NULL,  
MEM_BALANCE DECIMAL(6,2) DEFAULT 0);  
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [EliteVideo]> CREATE TABLE RENTAL (RENT_NUM INT PRIMARY KEY,  
RENT_DATE DATE NOT NULL,  
MEM_NUM INT NOT NULL,  
FOREIGN KEY(MEM_NUM) REFERENCES MEMBERSHIP(MEM_NUM));  
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [EliteVideo]> CREATE TABLE DETAILRENTAL (RENT_NUM INT NOT NULL,  
VID_NUM INT NOT NULL,  
DETAIL_FEE DECIMAL(6,2) NOT NULL,  
DETAIL_DUEDATE DATE NOT NULL,  
DETAIL_RETURNDATE DATE,  
DETAIL_DAILYLATEFEE DECIMAL(6,2),  
PRIMARY KEY (RENT_NUM, VID_NUM),
```

```
FOREIGN KEY(RENT_NUM) REFERENCES RENTAL(RENT_NUM),
FOREIGN KEY(VID_NUM) REFERENCES VIDEO(VID_NUM));
Query OK, 0 rows affected (0.009 sec)
```

Question 2

```
MariaDB [EliteVideo]> INSERT INTO MEMBERSHIP (MEM_NUM, MEM_FNAME, MEM_LNAME,
MEM_STREET, MEM_CITY, MEM_STATE, MEM_ZIP, MEM_BALANCE) VALUES
(102, 'Tami', 'Dawson', '2632 Takli Circle', 'Norene', 'TN', '37136', 11),
(103, 'Curt', 'Knight', '4025 Cornell Court', 'Flatgap', 'KY', '41219', 6),
(104, 'Jamal', 'Melendez', '788 East 145th Avenue', 'Quebeck', 'TN', '38579', 0),
(105, 'Iva', 'Mcclain', '6045 Musket Ball Circle', 'Summit', 'KY', '42783', 15),
(106, 'Miranda', 'Parks', '4469 Maxwell Place', 'Germantown', 'TN', '38183', 0),
(107, 'Rosario', 'Elliot', '7578 Danner Avenue', 'Columbia', 'TN', '38402', 5),
(108, 'Mattie', 'Guy', '4390 Evergreen Strret', 'Lily', 'KY', '40740', 0),
(109, 'Clint', 'Ochoa', '1711 Elm Street', 'Greenville', 'TN', '37745', 10),
(110, 'Lewis', 'Rosales', '4524 Southwind Circle', 'Counce', 'TN', '38326', 0),
(111, 'Stacy', 'Mann', '2789 East Cook Avenue', 'Murfreesboro', 'TN', '37132', 8),
(112, 'Luis', 'Trujillo', '7267 Melvin Avenue', 'Heiskel', 'TN', '37754', 3),
(113, 'Minnie', 'Gonzales', '6430 Vasili Drive', 'Williston', 'TN', '38076', 0);
Query OK, 12 rows affected (0.014 sec)
Records: 12 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> INSERT INTO RENTAL (RENT_NUM, RENT_DATE, MEM_NUM)
VALUES (1001, STR_TO_DATE('01-MAR-16', '%d-%b-%y'), 103),
(1002, STR_TO_DATE('01-MAR-16', '%d-%b-%y'), 105),
(1003, STR_TO_DATE('01-MAR-16', '%d-%b-%y'), 102),
(1004, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 110),
(1005, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 111),
(1006, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 107),
(1007, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 104),
(1008, STR_TO_DATE('03-MAR-16', '%d-%b-%y'), 105),
(1009, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 111);
Query OK, 8 rows affected (0.002 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
INSERT INTO PRICE (PRICE_CODE, PRICE_DESCRIPTION, PRICE_RENTFEE,
PRICE_DAILYLATEFEE)
VALUES
(1, 'Standard', 2.00, 1.00),
(2, 'New Release', 3.50, 3.00),
(3, 'Discount', 1.50, 1.00),
(4, 'Weekly Special', 1.00, 0.50);
Query OK, 4 rows affected (0.001 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
INSERT INTO MOVIE (MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR, MOVIE_COST, MOVIE_GENRE,
PRICE_CODE)
VALUES
(1234, 'The Cesar Family Christmas', 2014, 39.95, 'FAMILY', 2),
```

```
(1235, 'Smokey Mountain Wildlife', 2011, 59.95, 'ACTION', 1),
(1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2),
(1237, 'Beatnik Fever', 2014, 29.95, 'COMEDY', 2),
(1238, 'Constant Companion', 2015, 89.95, 'DRAMA', 3),
(1239, 'Where Hope Dies', 2005, 25.49, 'DRAMA', 3),
(1245, 'Time to Burn', 2012, 45.49, 'ACTION', 1),
(1246, 'What He Doesn't Know', 2013, 58.29, 'COMEDY', 1);
```

Query OK, 8 rows affected (0.001 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
INSERT INTO VIDEO (VID_NUM, VID_INDATE, MOVIE_NUM) VALUES (54321,
STR_TO_DATE('18-JUN-15', '%d-%b-%y'), 1234), (54324, STR_TO_DATE('18-JUN-15',
'%d-%b-%y'), 1234), (54325, STR_TO_DATE('18-JUN-15', '%d-%b-%y'), 1234), (34341,
STR_TO_DATE('22-JAN-14', '%d-%b-%y'), 1235), (34342, STR_TO_DATE('22-JAN-14',
'%d-%b-%y'), 1235), (34366, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 1236), (34367,
STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 1236), (34368, STR_TO_DATE('02-MAR-16',
'%d-%b-%y'), 1236), (34369, STR_TO_DATE('02-MAR-16', '%d-%b-%y'), 1236), (44392,
STR_TO_DATE('21-OCT-15', '%d-%b-%y'), 1237), (44397, STR_TO_DATE('21-OCT-15',
'%d-%b-%y'), 1237), (59239, STR_TO_DATE('14-FEB-16', '%d-%b-%y'), 1237), (61388,
STR_TO_DATE('25-JAN-14', '%d-%b-%y'), 1239), (61353, STR_TO_DATE('28-JAN-13',
'%d-%b-%y'), 1245), (61354, STR_TO_DATE('28-JAN-13', '%d-%b-%y'), 1245), (61367,
STR_TO_DATE('30-JUL-15', '%d-%b-%y'), 1246), (61369, STR_TO_DATE('30-JUL-15',
'%d-%b-%y'), 1246);
```

Query OK, 17 rows affected (0.001 sec)

Records: 17 Duplicates: 0 Warnings: 0

```
INSERT INTO DETAILRENTAL (RENT_NUM, VID_NUM, DETAIL_FEE, DETAIL_DUEDATE,
DETAIL_RETURNDATE, DETAIL_DAILYLATEFEE) VALUES (1001, 34342, 2,
STR_TO_DATE('04-MAR-16', '%d-%b-%y'), STR_TO_DATE('02-MAR-16', '%d-%b-%y'), NULL);
Query OK, 1 row affected (0.002 sec)
```

Question 3

```
select MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE where MOVIE_NUM = 1245;
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_YEAR
1245	Time to Burn	2012

1 row in set (0.005 sec)

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE MOVIE_NUM = 1245;
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_YEAR
1245	Time to Burn	2010

1 row in set (0.000 sec)

Question 4

```
select movie_num, movie_title, movie_genre, price_code from movie where movie_genre = 'ACTION';
```

movie_num	movie_title	movie_genre	price_code
1235	Smokey Mountain Wildlife	ACTION	1
1245	Time to Burn	ACTION	1

2 rows in set (0.001 sec)

```
select movie_num, movie_title, movie_genre, price_code from movie where movie_genre = 'ACTION';
```

movie_num	movie_title	movie_genre	price_code
1235	Smokey Mountain Wildlife	ACTION	3
1245	Time to Burn	ACTION	3

2 rows in set (0.001 sec)

Question 5

```
select price_code, price_description, price_rentfee from price;
```

price_code	price_description	price_rentfee
1	Standard	2.00
2	New Release	3.50
3	Discount	1.50
4	Weekly Special	1.00

4 rows in set (0.001 sec)

```
update price set price_rentfee = price_rentfee + 0.50;
```

Query OK, 4 rows affected (0.005 sec)

Rows matched: 4 Changed: 4 Warnings: 0

```
MariaDB [elitevideo]> select price_code, price_description, price_rentfee from price;
```

price_code	price_description	price_rentfee
1	Standard	2.50
2	New Release	4.00
3	Discount	2.00
4	Weekly Special	1.50

Question 6.

MariaDB [elitevideo]> select * from movie;

MOVIE_NUM PRICE_CODE	MOVIE_TITLE	MOVIE_YEAR	MOVIE_COST	MOVIE_GENRE
1234 2	The Cesar Family Christmas	2014	39.95	FAMILY
1235 3	Smokey Mountain Wildlife	2011	59.95	ACTION
1236 2	Richard Goodhope	2015	59.95	DRAMA
1237 2	Beatnik Fever	2014	29.95	COMEDY
1238 3	Constant Companion	2015	89.95	DRAMA
1239 3	Where Hope Dies	2005	25.49	DRAMA
1245 3	Time to Burn	2010	45.49	ACTION
1246 1	What He Doesn't Know	2013	58.29	COMEDY

8 rows in set (0.007 sec)

Question 7

MariaDB [elitevideo]> select MOVIE_YEAR, MOVIE_TITLE, MOVIE_COST from movie ORDER by movie_year desc;

MOVIE_YEAR	MOVIE_TITLE	MOVIE_COST
2015	Richard Goodhope	59.95
2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesn't Know	58.29
2011	Smokey Mountain Wildlife	59.95
2010	Time to Burn	45.49
2005	Where Hope Dies	25.49

8 rows in set (0.004 sec)

Question 8

MariaDB [elitevideo]> select movie_title, movie_year, movie_genre from movie order by movie_genre asc, movie_year desc;

movie_title	movie_year	movie_genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION
Beatnik Fever	2014	COMEDY
What He Doesn't Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
The Cesar Family Christmas	2014	FAMILY

8 rows in set (0.003 sec)

Question 9.

MariaDB [elitevideo]> select movie_num, movie_title, price_code from movie where movie_title like 'R%';

movie_num	movie_title	price_code
1236	Richard Goodhope	2

1 row in set (0.006 sec)

Question 10.

MariaDB [elitevideo]> select movie_title, movie_year, movie_cost from movie where movie_title like '%hope%' order by movie_title asc;

movie_title	movie_year	movie_cost
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

2 rows in set (0.001 sec)

Question 11

MariaDB [elitevideo]> select movie_title, movie_year, movie_genre from movie where movie_genre = 'action';

movie_title	movie_year	movie_genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION

2 rows in set (0.002 sec)

Question 12

```
MariaDB [elitevideo]> select movie_num, movie_title, movie_cost from movie where
movie_cost >40;
```

movie_num	movie_title	movie_cost
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49
1246	What He Doesn't Know	58.29

5 rows in set (0.002 sec)

Question 13

```
MariaDB [elitevideo]> select movie_num, movie_title, movie_cost, movie_genre from
movie where (movie_genre = 'action' or movie_genre = 'comedy') and movie_cost <50
order by movie_genre asc;
```

movie_num	movie_title	movie_cost	movie_genre
1245	Time to Burn	45.49	ACTION
1237	Beatnik Fever	29.95	COMEDY

2 rows in set (0.001 sec)

Question 14

```
MariaDB [elitevideo]> select mem_num, mem_fname, mem_lname, mem_street, mem_state,
mem_balance from membership where mem_state = 'TN' and mem_balance <5 and
mem_street like '%Avenue';
```

mem_num	mem_fname	mem_lname	mem_street	mem_state	mem_balance
104	Jamal	Melendez	788 East 145th Avenue	TN	0.00
112	Luis	Trujillo	7267 Melvin Avenue	TN	3.00

2 rows in set (0.001 sec)

Question 15

```
MariaDB [elitevideo]> select movie_genre, count(*) as num_movie from movie group by
movie_genre;
```

```

+-----+-----+
| movie_genre | num_movie |
+-----+-----+
| ACTION      |          2 |
| COMEDY      |          2 |
| DRAMA       |          3 |
| FAMILY      |          1 |
+-----+-----+
4 rows in set (0.005 sec)

```

Question 16

```

MariaDB [elitevideo]> select Round(avg(movie_cost), 4) as avg_cost from movie;
+-----+
| avg_cost |
+-----+
| 51.1275 |
+-----+
1 row in set (0.005 sec)

```

Question 17

```

MariaDB [elitevideo]> select movie_genre, round(avg(movie_cost), 4) as avg_cost
from movie group by movie_genre;
+-----+-----+
| movie_genre | avg_cost |
+-----+-----+
| ACTION      | 52.7200 |
| COMEDY      | 44.1200 |
| DRAMA       | 58.4633 |
| FAMILY      | 39.9500 |
+-----+-----+
4 rows in set (0.005 sec)

```

Question 18

```

MariaDB [elitevideo]> select m.movie_title, m.movie_genre, p.price_description,
price_rentfee from movie m join price p on m.price_code = p.price_code;
+-----+-----+-----+-----+
| movie_title          | movie_genre | price_description | price_rentfee |
+-----+-----+-----+-----+
| The Cesar Family Christmas | FAMILY      | New Release      | 4.00          |
| Smokey Mountain Wildlife  | ACTION      | Discount          | 2.00          |
| Richard Goodhope         | DRAMA       | New Release      | 4.00          |
| Beatnik Fever            | COMEDY      | New Release      | 4.00          |
| Constant Companion       | DRAMA       | Discount          | 2.00          |
| Where Hope Dies          | DRAMA       | Discount          | 2.00          |
| Time to Burn             | ACTION      | Discount          | 2.00          |
| What He Doesn't Know     | COMEDY      | Standard          | 2.50          |
+-----+-----+-----+-----+

```


8 rows in set (0.005 sec)

Question 19

```
MariaDB [elitevideo]> select m.movie_genre, round(avg(p.price_rentfee), 4) as  
avg_rentfee from movie m join price p on m.price_code = p.price_code group by  
m.movie_genre;
```

movie_genre	avg_rentfee
ACTION	2.0000
COMEDY	3.2500
DRAMA	2.6667
FAMILY	4.0000

4 rows in set (0.006 sec)

Question 20

```
MariaDB [elitevideo]> select m.movie_title, round(m.movie_cost/p.price_rentfee, 2)  
as Break_even_rentals from movie m join price p on m.price_code = p.price_code;
```

movie_title	Break_even_rentals
The Cesar Family Christmas	9.99
Smokey Mountain Wildlife	29.98
Richard Goodhope	14.99
Beatnik Fever	7.49
Constant Companion	44.98
Where Hope Dies	12.75
Time to Burn	22.75
What He Doesn't Know	23.32

8 rows in set (0.001 sec)

Question 21.

```
MariaDB [elitevideo]> select movie_title, movie_year from movie where price_code is  
not null;
```

movie_title	movie_year
The Cesar Family Christmas	2014
Smokey Mountain Wildlife	2011
Richard Goodhope	2015
Beatnik Fever	2014
Constant Companion	2015
Where Hope Dies	2005
Time to Burn	2010
What He Doesn't Know	2013

```
+-----+-----+
8 rows in set (0.005 sec)
```

Question 22

```
MariaDB [elitevideo]> select movie_title, movie_genre, movie_cost from movie where
movie_cost between 44.99 and 49.99;
```

```
+-----+-----+-----+
| movie_title | movie_genre | movie_cost |
+-----+-----+-----+
| Time to Burn | ACTION      | 45.49      |
+-----+-----+-----+
```

```
1 row in set (0.001 sec)
```

Question 23.

```
MariaDB [elitevideo]> select m.movie_title, p.price_description, p.price_rentfee
from movie m join price p on m.price_code = p.price_code where m.movie_genre in
('Family', 'Comedy', 'Drama');
```

```
+-----+-----+-----+
| movie_title          | price_description | price_rentfee |
+-----+-----+-----+
| The Cesar Family Christmas | New Release      | 4.00          |
| Richard Goodhope        | New Release      | 4.00          |
| Beatnik Fever           | New Release      | 4.00          |
| Constant Companion      | Discount         | 2.00          |
| Where Hope Dies         | Discount         | 2.00          |
| What He Doesn't Know    | Standard         | 2.50          |
+-----+-----+-----+
```

```
6 rows in set (0.001 sec)
```

Question 24

```
MariaDB [elitevideo]> select m.mem_num, m.mem_fname, m.mem_lname, m.mem_balance
from membership m join rental r on m.mem_num = r.mem_num;
```

```
+-----+-----+-----+-----+
| mem_num | mem_fname | mem_lname | mem_balance |
+-----+-----+-----+-----+
| 102     | Tami      | Dawson    | 11.00       |
| 103     | Curt      | Knight    | 6.00        |
| 104     | Jamal     | Melendez  | 0.00        |
| 105     | Iva       | McClain   | 15.00       |
| 105     | Iva       | McClain   | 15.00       |
| 107     | Rosario   | Elliot    | 5.00        |
| 110     | Lewis     | Rosales   | 0.00        |
| 111     | Stacy     | Mann      | 8.00        |
| 111     | Stacy     | Mann      | 8.00        |
+-----+-----+-----+-----+
```

```
9 rows in set (0.002 sec)
```

```
MariaDB [elitevideo]> select Distinct m.mem_num, m.mem_fname, m.mem_lname,
```

```
m.mem_balance from membership m join rental r on m.mem_num = r.mem_num;
```

mem_num	mem_fname	mem_lname	mem_balance
102	Tami	Dawson	11.00
103	Curt	Knight	6.00
104	Jamal	Melendez	0.00
105	Iva	Mcclain	15.00
107	Rosario	Elliot	5.00
110	Lewis	Rosales	0.00
111	Stacy	Mann	8.00

7 rows in set (0.001 sec)

Question 25

```
MariaDB [elitevideo]> select min(m.mem_balance) as min_balance, max(m.mem_balance) as max_balance, avg(m.mem_balance) as avg_balance from membership m where m.mem_num in (select Distinct mem_num from rental);
```

min_balance	max_balance	avg_balance
0.00	15.00	6.428571

1 row in set (0.006 sec)

Question 26

```
MariaDB [elitevideo]> select r.rent_num, r.rent_date, dr.Vid_num, m.movie_title, dr.detail_duedate, dr.detail_returndate from rental r join detailrental dr on r.rent_num = dr.rent_num join video v on dr.vid_num = v.vid_num join movie m on v.movie_num = m.movie_num where dr.detail_returndate > dr.detail_duedate order by r.rent_num asc, m.movie_title asc;
```

Empty set (0.002 sec)

Question 27.

```
MariaDB [elitevideo]> select r.rent_num, r.rent_date, m.movie_title, detail_fee from rental r join detailrental dr on r.rent_num = dr.rent_num join video v on dr.vid_num = v.vid_num join movie m on v.movie_num = m.movie_num where dr.detail_returndate <= dr.detail_duedate order by r.rent_num, m.movie_title;
```

rent_num	rent_date	movie_title	detail_fee
1001	2016-03-01	Smokey Mountain Wildlife	2.00

1 row in set (0.001 sec)

Question 28

```
ariaDB [elitevideo]> select movie_num, movie_genre, AVG(movie_cost) over (partition
by movie_genre) as AVG_GENRE_COST, movie_cost, ((movie_cost - AVG(movie_cost) over
(partition by movie_genre)) / AVG(Movie_cost) over (partition by movie_genre)) *
100 as PERCENT_DIFF from movie;
```

movie_num	movie_genre	AVG_GENRE_COST	movie_cost	PERCENT_DIFF
1235	ACTION	52.720000	59.95	13.7139605463
1245	ACTION	52.720000	45.49	-13.7139605463
1237	COMEDY	44.120000	29.95	-32.1169537625
1246	COMEDY	44.120000	58.29	32.1169537625
1236	DRAMA	58.463333	59.95	2.5429043851
1238	DRAMA	58.463333	89.95	53.8571184227
1239	DRAMA	58.463333	25.49	-56.4000228061
1234	FAMILY	39.950000	39.95	0.0000000000

8 rows in set (0.001 sec)